Simple Linear Regression

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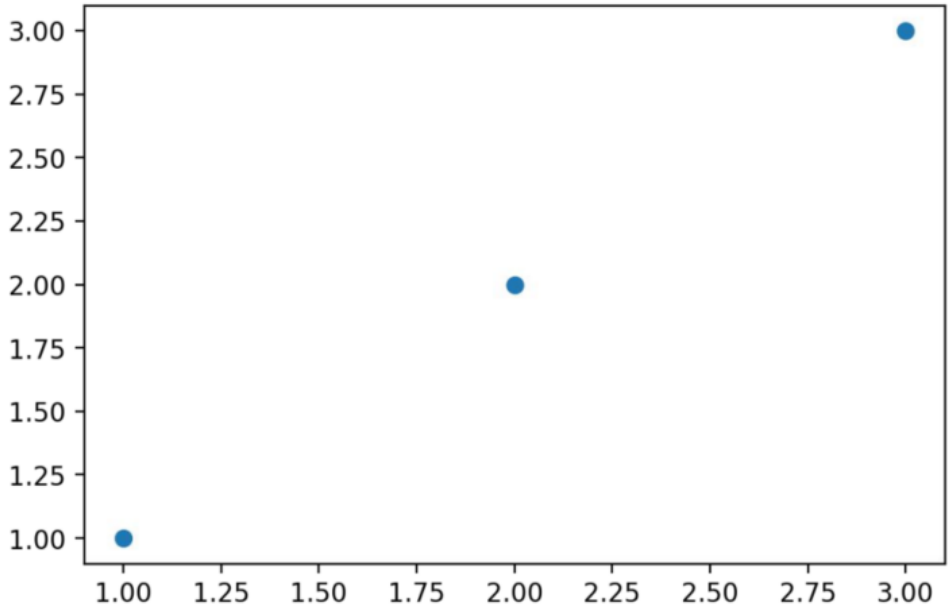
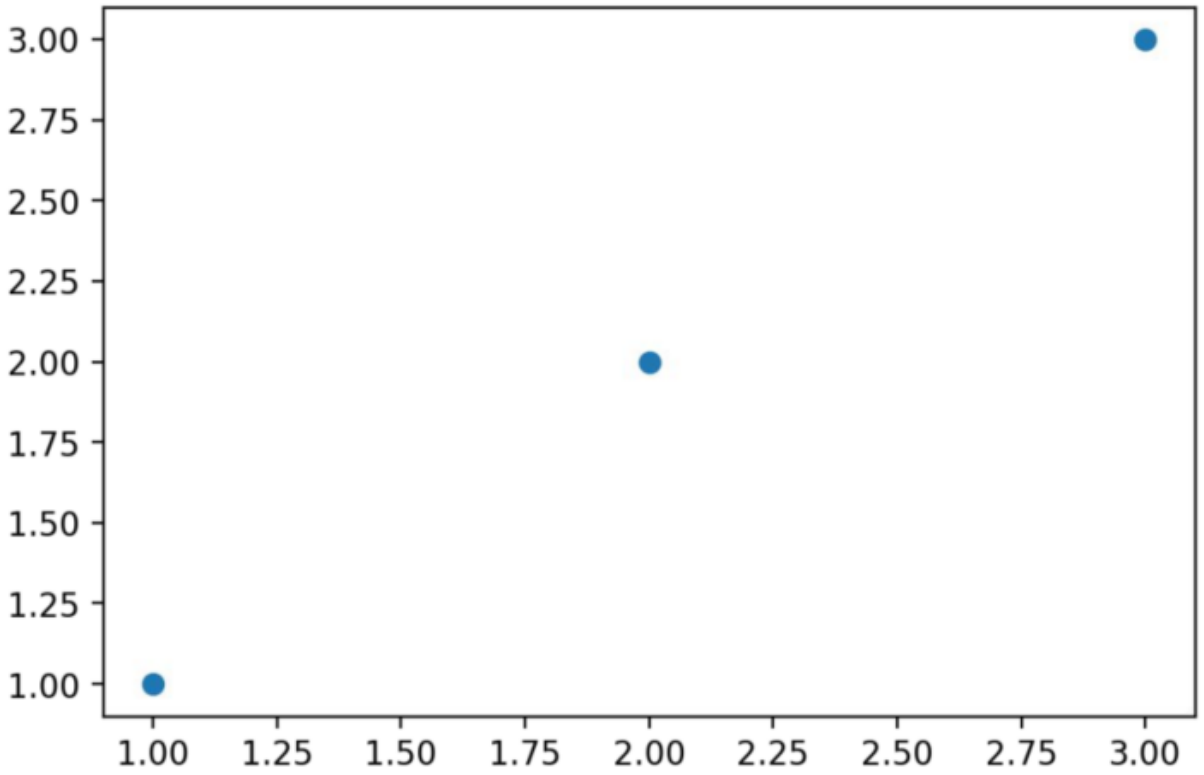
Linear Relationship

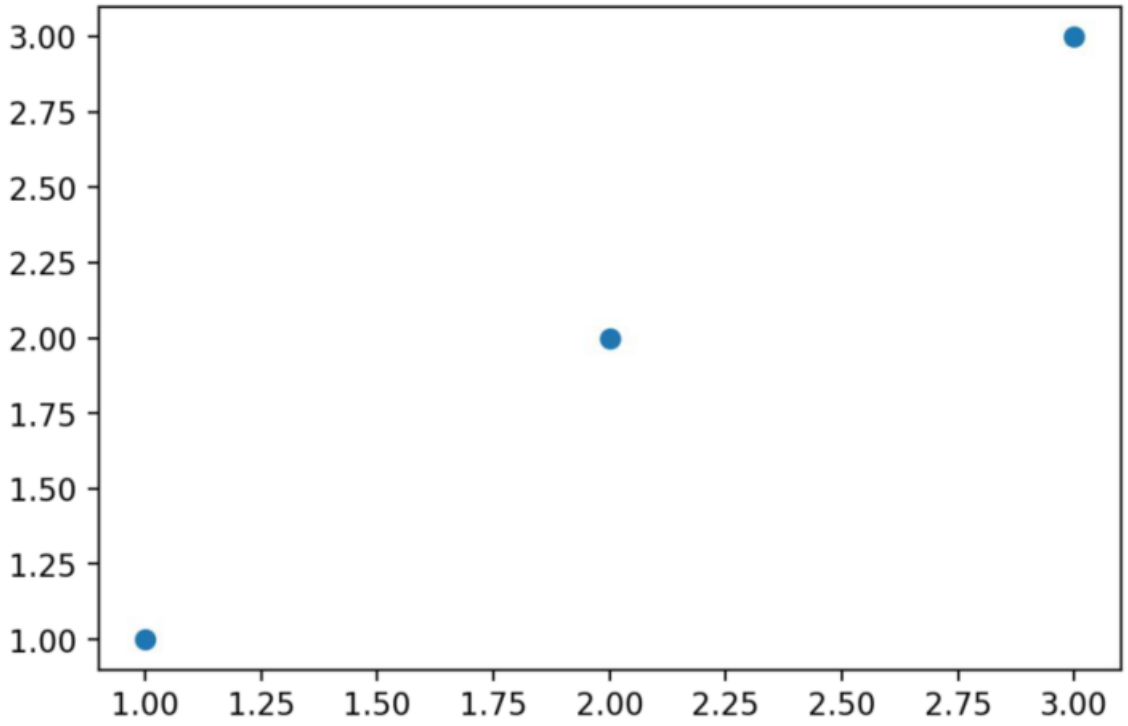
- implies some constant straight-line relationship
- simplest possible being y = x



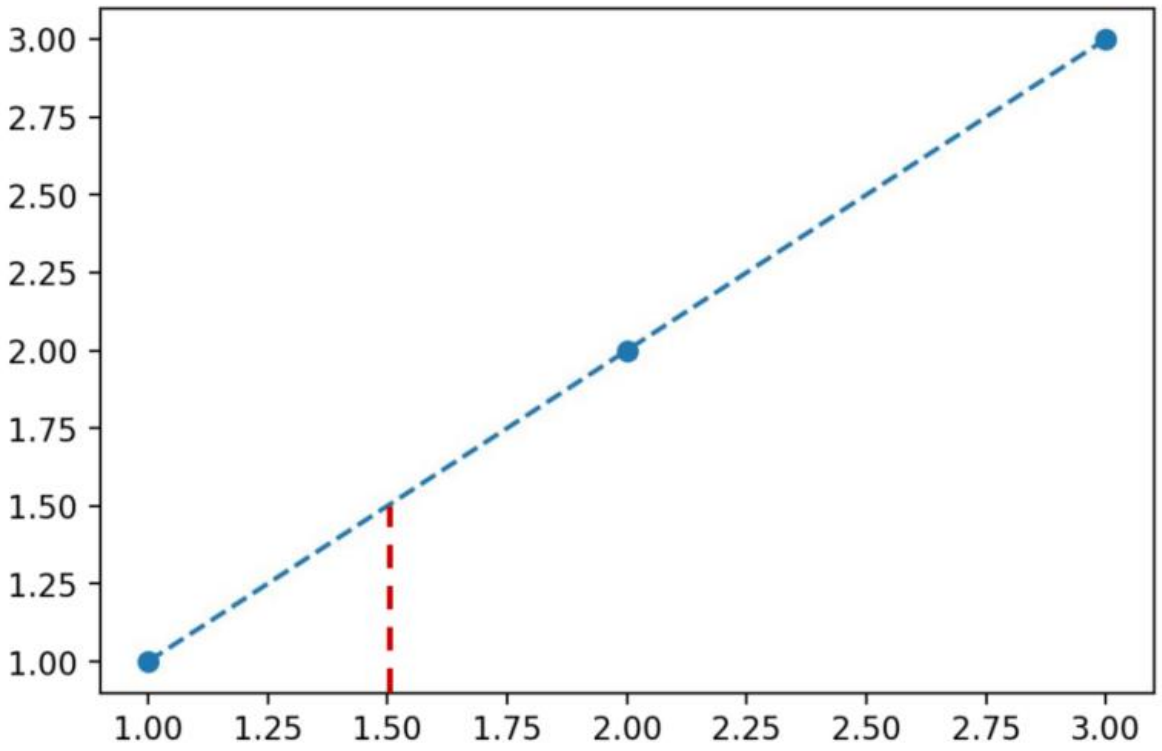
Here we see x = [1, 2, 3] and y = [1, 2, 3]



We could build out the relationship y = x as our "fitted" line



This implies for some new x value I can predict it is already related to y



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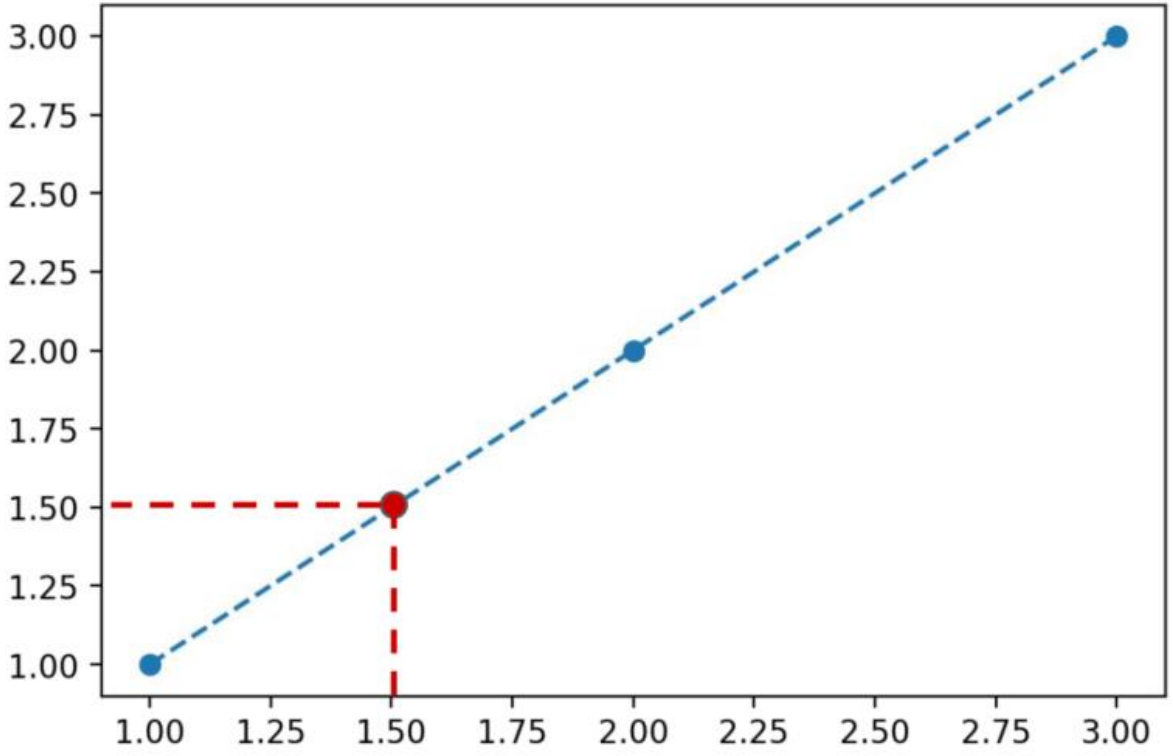
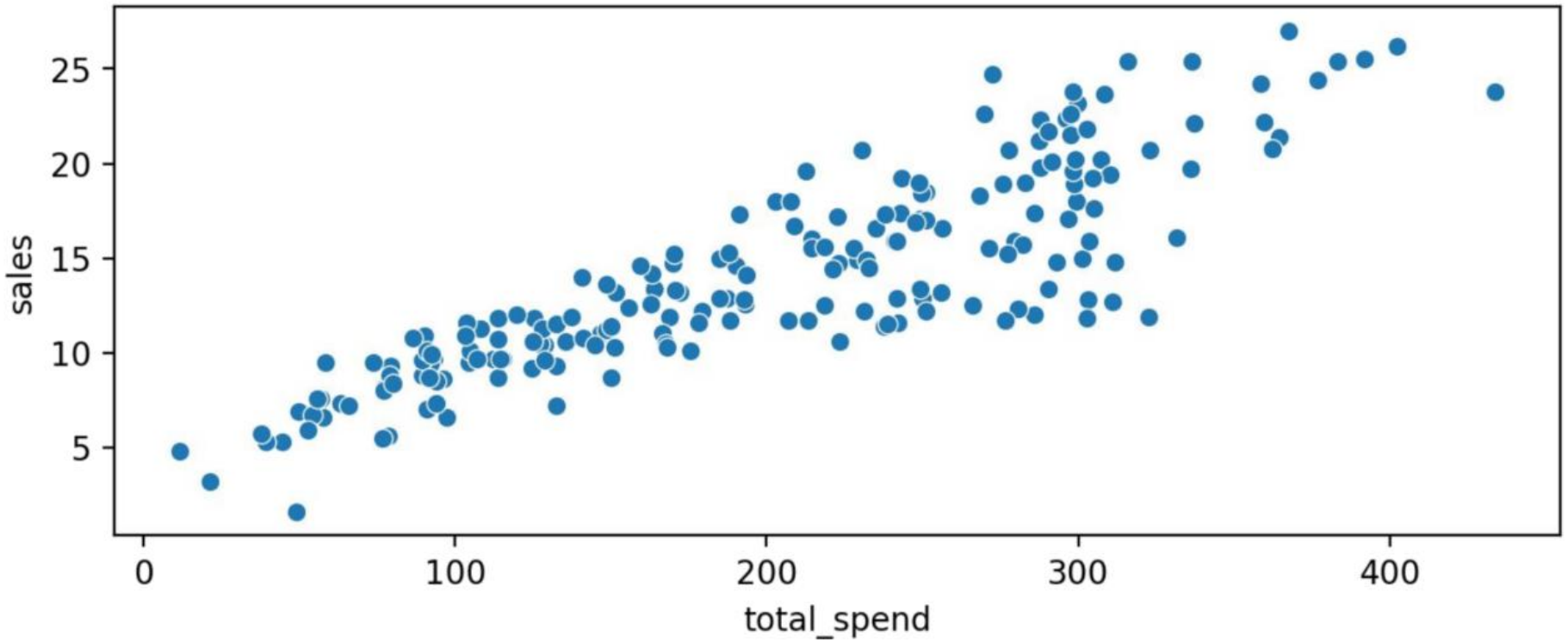


But what happens with real data? Where do we draw the line for this?



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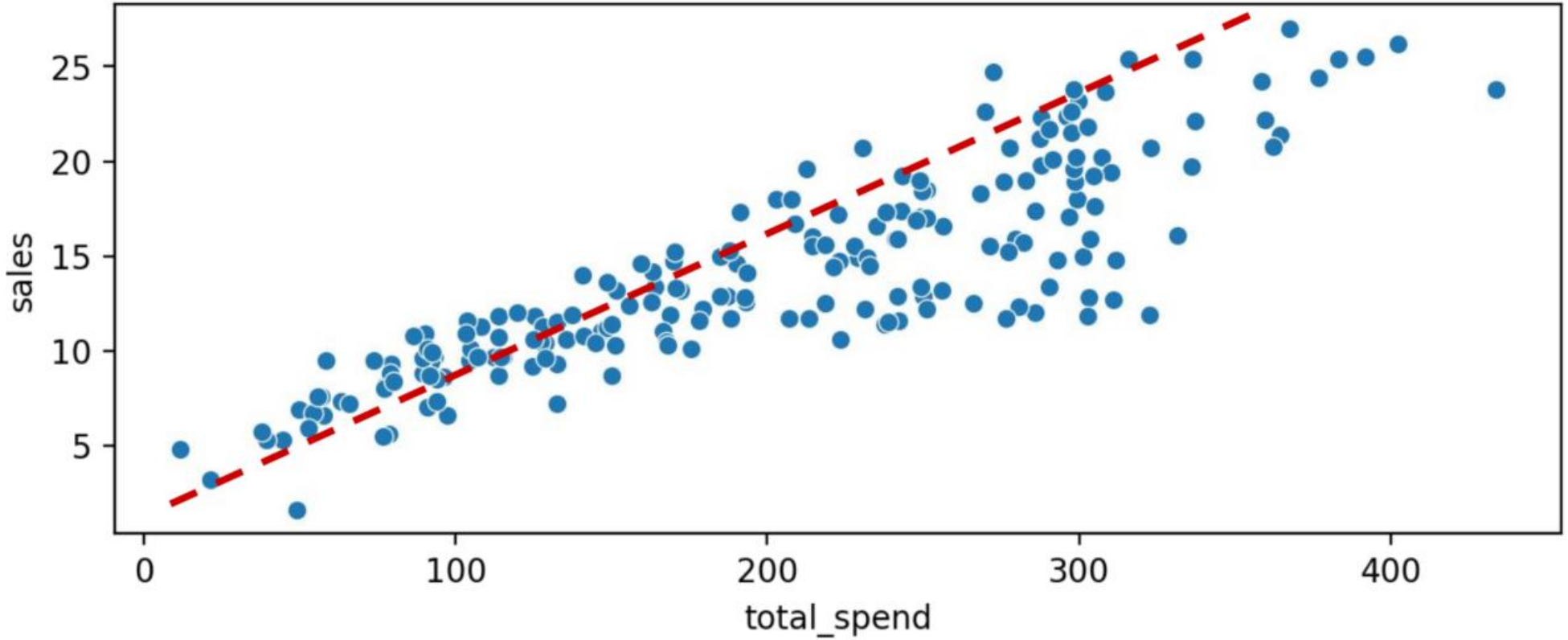

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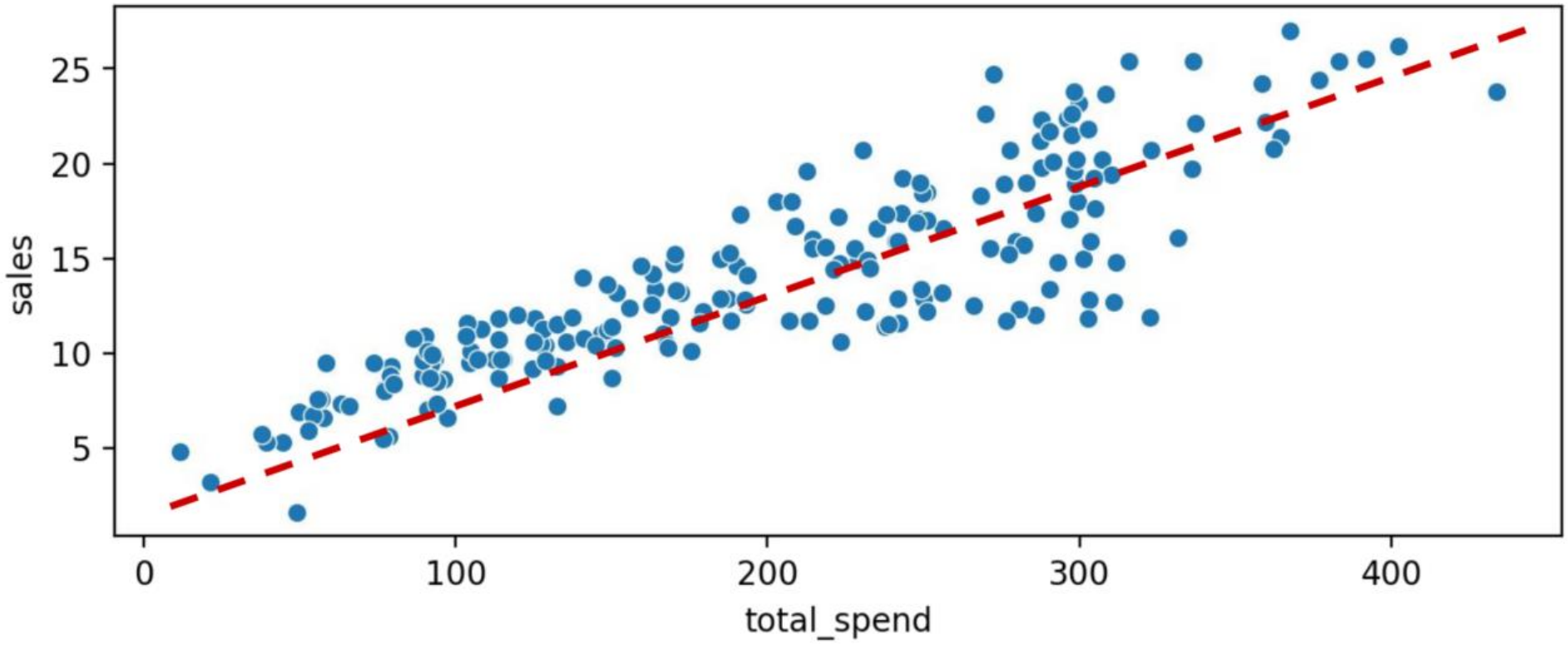
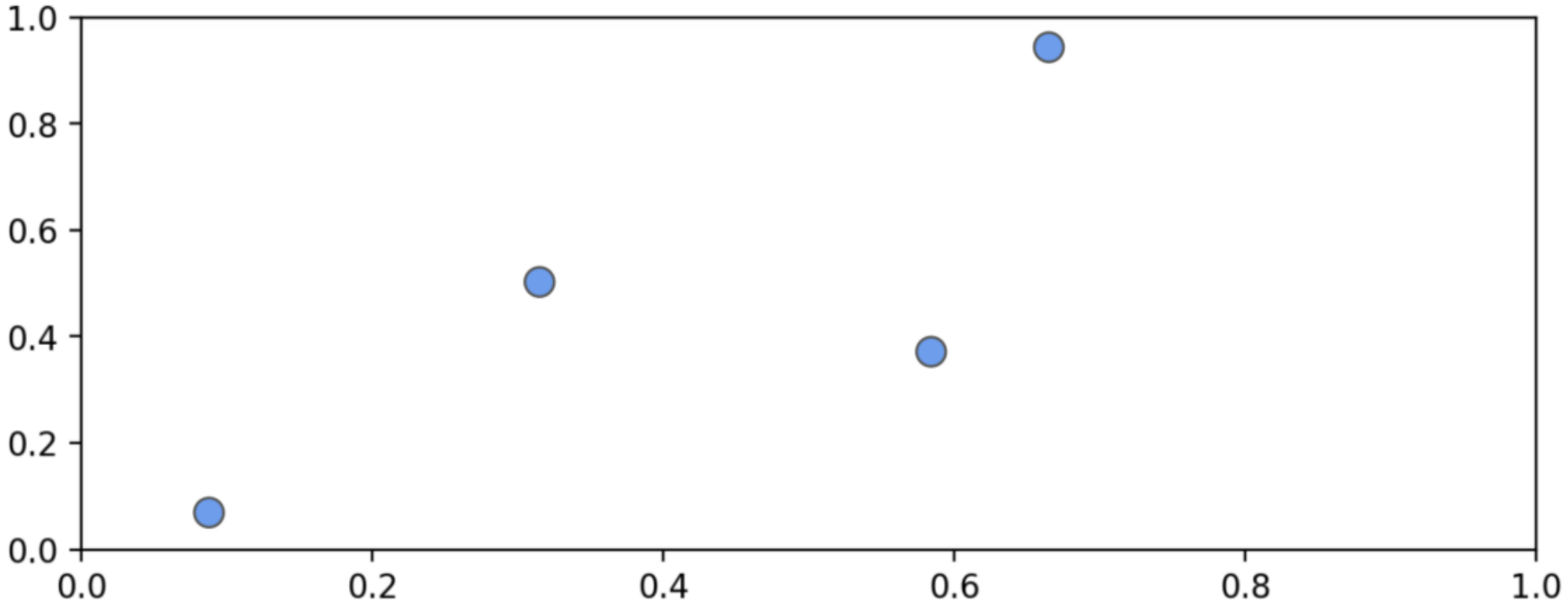


We understand we want to minimize the overall distance from the points to the line



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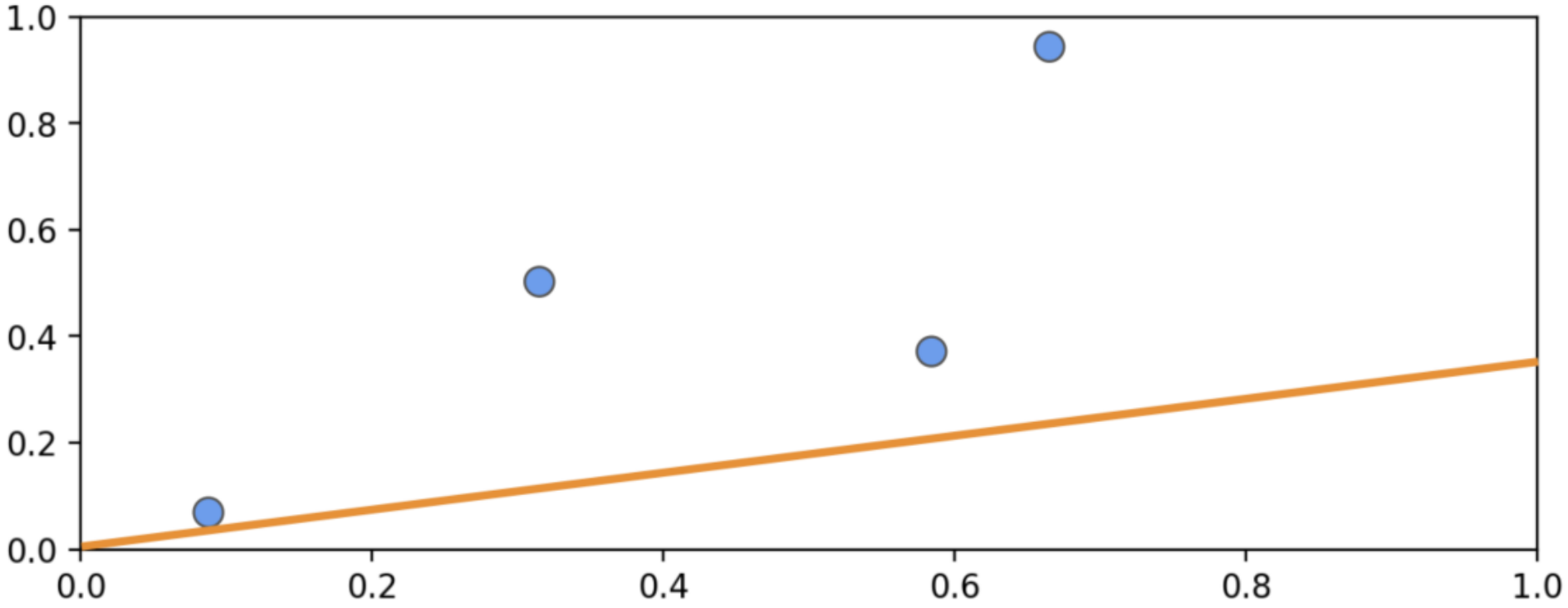
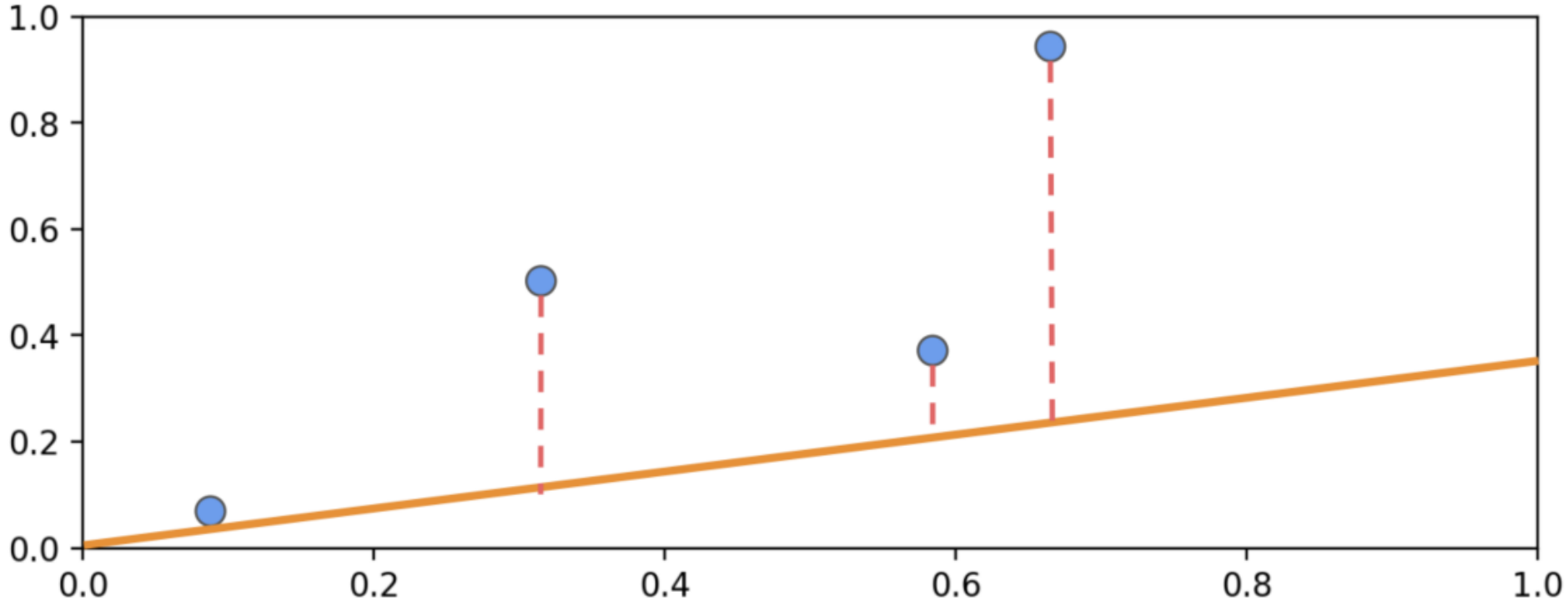
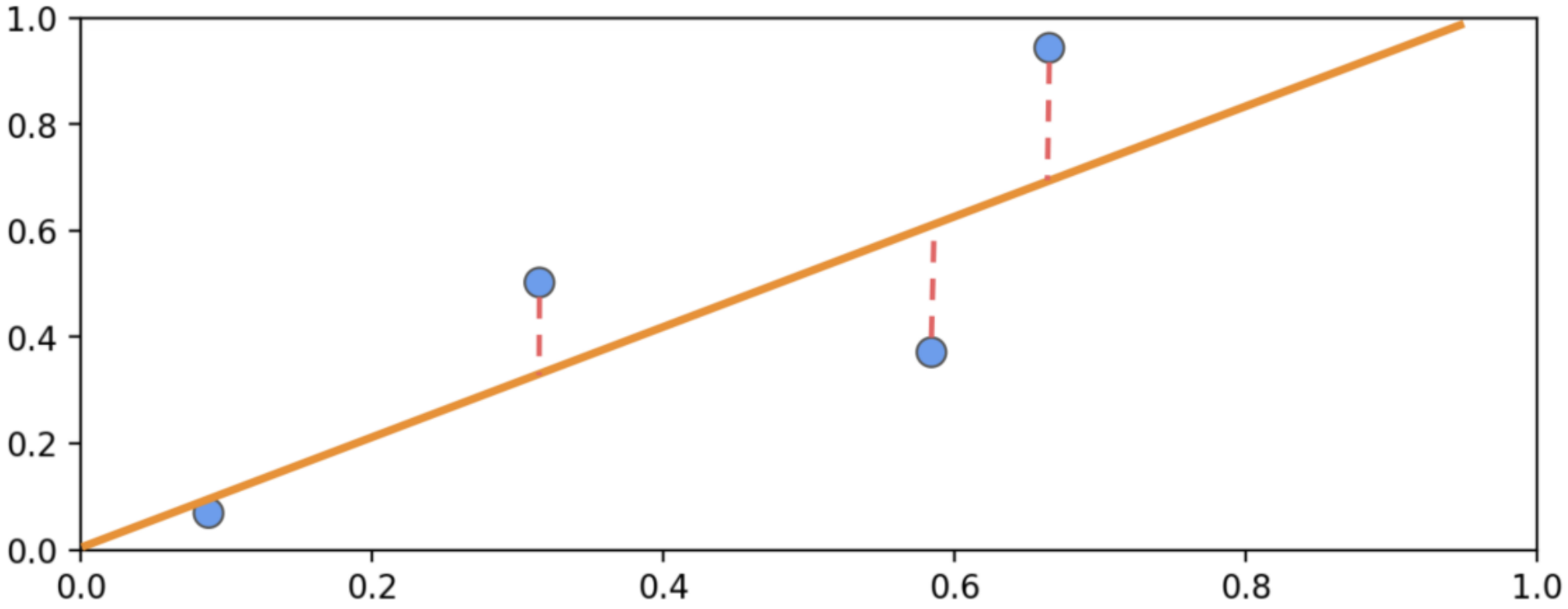


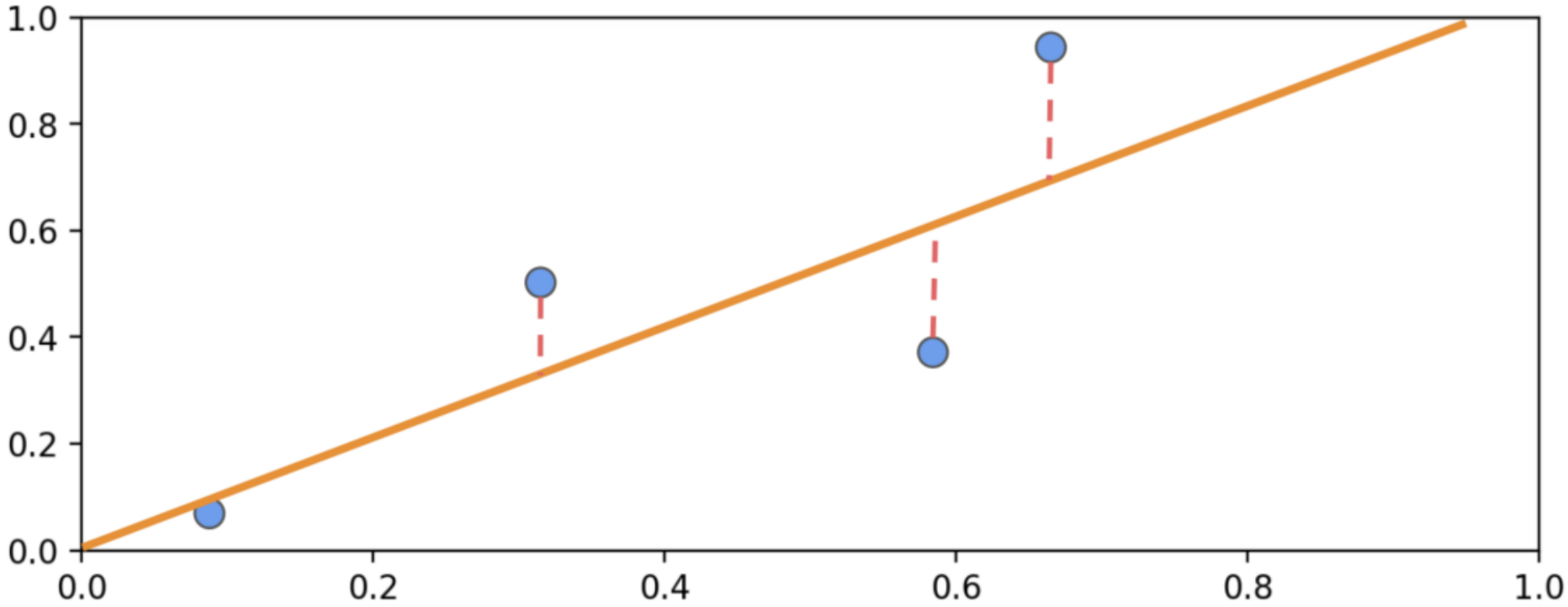
We also know we can measure this error from the real data points to the line, the residual error



Some lines will clearly be better fits than others

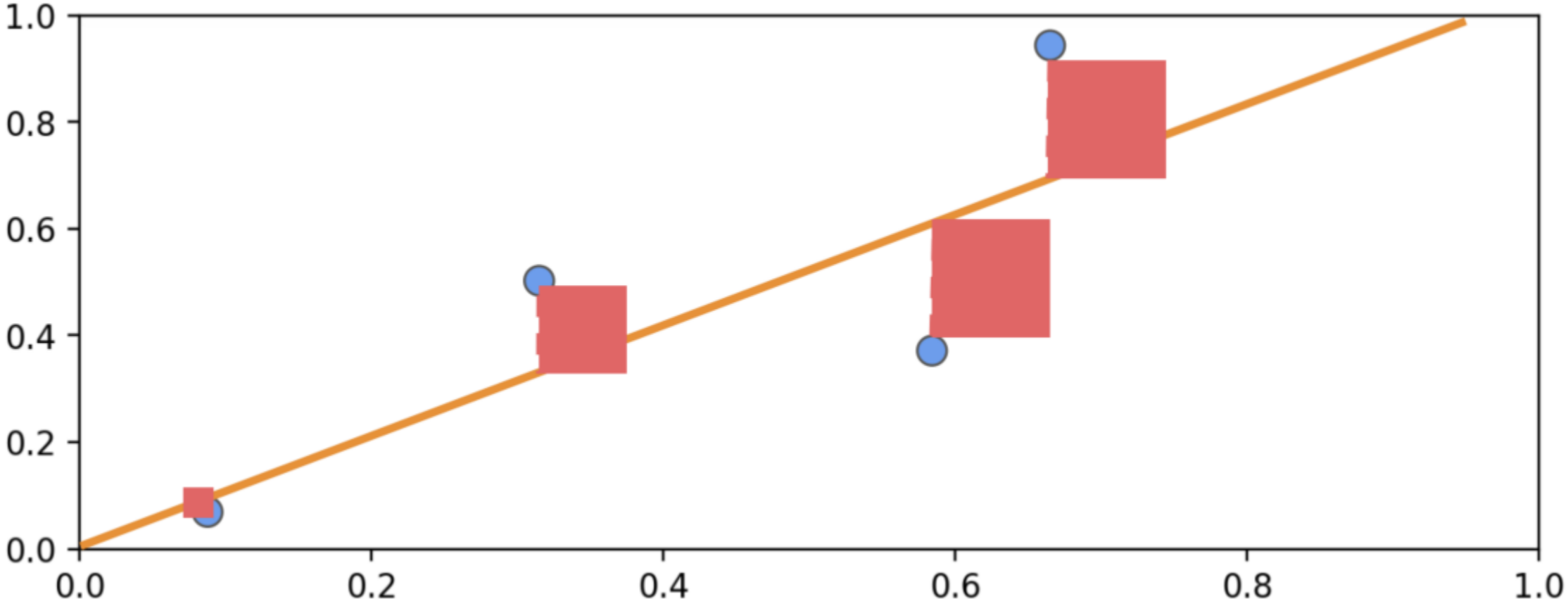


We can also see the residuals can be both positive and negative



Ordinary least squares (OLS) works by minimizing the sum of the squares of the differences between observed dependent variable in the given dataset and those predicted by the linear function

We can visualize squared error to minimize:



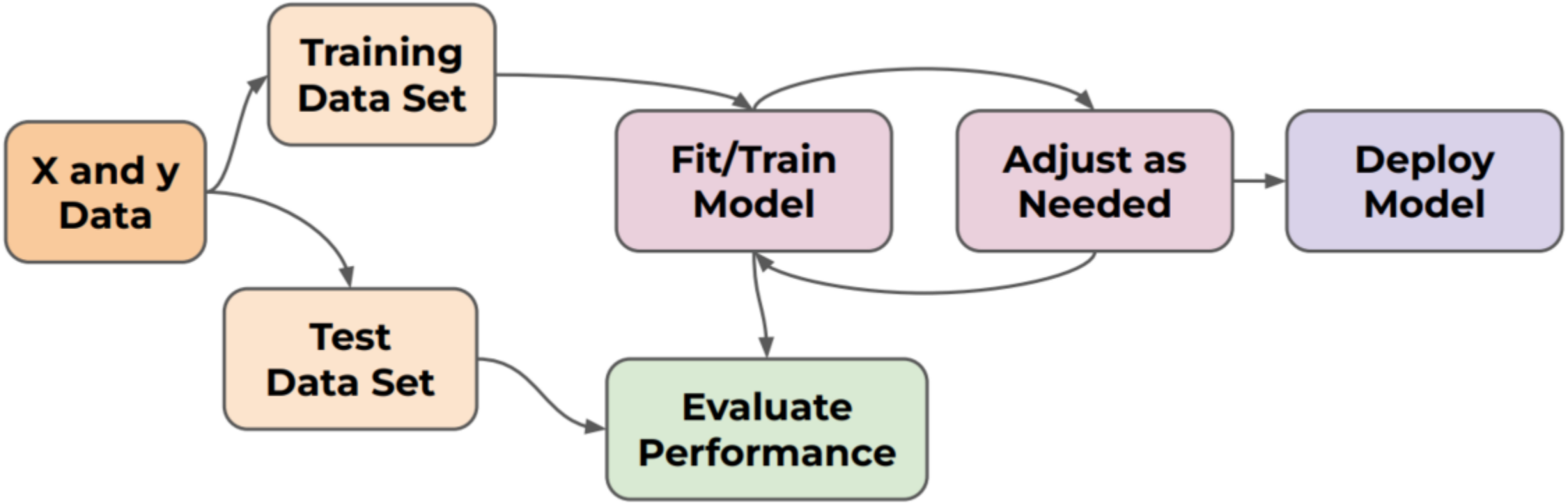
A simple straight line is y = mx + b

- where m is the slope
- b is the intercept with y-axis

We can see for y = mx + b, there is only room for one possible feature x

OLS will allow us to directly solve for the slope m and intercept b.

Recall and understand the process of machine learning modeling



Thank you very much for listening.